DES Pune University, Pune
School of Engineering and Technology
Department of Computer Engineering and Technology
Program: B. Tech in Computer Science and Engineering

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Subject: Python Programming		
Assignment No.: 7		
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Lab Assignment: 07

Aim:
To write a Python program to calculate the roots of a quadratic equation.

Objective:
To accept coefficients of a quadratic equation and compute its roots using the discriminant method.

Theory:
A quadratic equation is of the form

ax^2+bx+c=0

The nature of roots depends on the discriminant

D=b^2−4ac

- If D>0: Two distinct real roots
- If D=0: One real root
- If D<0: No real roots

Algorithm:
1. Start
2. Read values of a, b, and c
3. Calculate discriminant
4. If D > 0, calculate roots
5. Elif D=0, calculate roots
6. Else, display no real roots
7. Stop

Pseudo Code:
```
START
READ a, b, c
D = b*b - 4*a*c
IF D > 0
   r1 = (-b + sqrt(D)) / (2*a)
   r2 = (-b - sqrt(D)) / (2*a)
   PRINT r1, r2
```

ELIF
  r1 = (-b + sqrt(D)) / (2*a)
  r2 = (-b - sqrt(D)) / (2*a)
  PRINT r1, r2
ELSE
  PRINT "No real roots"
END IF
STOP

Code:

```
import math
a = float(input("Enter a: "));
b = float(input("Enter b: "));
c = float(input("Enter c: "));
d = b*b - 4*a*c;
if d > 0:
    r1 = (-b + math.sqrt(d)) / (2*a);
    r2 = (-b - math.sqrt(d)) / (2*a);
    print("Real and distinct roots are obtained. Roots are:", r1, "and", r2);
elif d==0:
    r1 = (-b + math.sqrt(d)) / (2*a);
    r2 = (-b - math.sqrt(d)) / (2*a);
    print("Real and equal roots are obtained. Roots are:", r1, "and", r2);
else:
    print("No real roots");
```

Output:

```
Enter a: 1
Enter b: -5
Enter c: 6
Real and distinct roots are obtained. Roots are: 3.0 and 2.0
```

```
Enter a: 1
Enter b: -4
Enter c: 4
Real and equal roots are obtained. Roots are: 2.0 and 2.0
```

```
Enter a: 4
Enter b: 7
Enter c: 8
No real roots
```

Conclusion:

The program successfully calculates the roots of a quadratic equation based on the discriminant value.